# Pedestrian Detection with 3 Types of Data Fusion

## 1. Introduction and Objective

This report analyzes and compares three different data fusion approaches applied to pedestrian detection under low-visibility conditions and in the presence of atmospheric disturbances (rain). The main objective is to evaluate whether using a neural network and applying a fine-tuning technique can improve pedestrian detection accuracy compared to a traditional non-learning-based fusion method.

## 2. Dataset and Experimental Setup

The study uses two types of images provided by the sensors:

- **Visible Image:** captured by a color camera.
- **Thermal Image:** captured by an infrared camera.

To replicate critical weather scenarios, artificial rain was added exclusively to the visible images during experiments, leaving the thermal images unchanged. The detection framework relies on a common **YOLO** model across all methods for the final extraction of pedestrian bounding boxes.

## 3. Comparison of the Three Data Fusion Methods

The processing pipeline merges the visible image (with rain) and the thermal image before feeding the output into the YOLO model:

1. **No Learning-Based Data Fusion (Fixed Average):**
   - **Rule:** Computes the pixel-by-pixel average of the two input images ($\frac{\text{Visible}+\text{Thermal}}{2}\frac{\text{Visible}+\text{Thermal}}{2}$).
   - **Characteristics:** Each pixel receives a fixed 50% contribution. Weights are uniform across the entire scene, and the system involves no learning phase.
2. **Learning-Based Data Fusion (Pre-trained AMG-Fuse):**
   - **Model:** Uses the AMG-Fuse network pre-trained on the AWMM-100K dataset (1,000 image pairs).
   - **Operation:** Extracts features, combines them via attention mechanisms focusing on edges and contrasts, and produces a combined image $RR$. The contribution of each sensor is dynamically weighted using a spatial coefficient $\alpha\alpha$ calculated based on the scene ($\alpha > 0.5\alpha > 0.5$ favors visible, $\alpha < 0.5\alpha < 0.5$ favors thermal).
3. **Fine-Tuned Learning-Based Data Fusion:**
   - **Model:** Starts from the same AMG-Fuse network and applies targeted training (*fine-tuning*) limited only to the network's final layer.
   - **Dataset & Training:** Uses 100 image pairs from the LLVIP dataset over 3 epochs, updating 2,592 weights based on the detection error output by YOLO compared to ground truth annotations.

## 4. Results and Performance Evaluation

The three methods were evaluated on a test set of **50 image pairs from the LLVIP dataset** (containing 128 annotated pedestrians), using the exact same frozen YOLO model.
The pedestrian detection counts revealed the following metrics:

- **Pedestrians Detected:**
  - Fixed average: 67
  - Pre-trained fusion: 73
  - Fine-tuned fusion: **84**
- **Pedestrians Missed:**
  - Fixed average: 61
  - Pre-trained fusion: 55

    - Fine-tuned fusion: **44**
- **False Positives:**
    - Fixed average: 16
    - Pre-trained fusion: **11**
    - Fine-tuned fusion: 17

# 5. Conclusions

The comparative analysis demonstrates that applying neural network-based data fusion, and particularly optimizing it through fine-tuning, significantly increases the number of correctly identified pedestrians (from 67 up to 84 detections) and reduces missed detections (from 61 down to 44). Although pure pre-trained fusion maintains a slightly lower false positive rate, target-domain fine-tuning yields the best overall performance in adverse rain conditions.